FPGA Hardware Accelerators for Drone Embedded AI

Feasibility Assessment, State-of-the-Art Survey & Thesis Proposal Framework

Research Date: July 22, 2026 **Depth:** Deep — Academic + Industry + Technical Architecture **Sources Consulted:** 20+ peer-reviewed papers, datasheets, academic surveys, industry benchmarks **Prepared for:** Master's Thesis in Computer Engineering — Embedded AI & Hardware Acceleration

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Executive Summary

The core proposal — using FPGA-based hardware accelerators on single-to-triple application drones to reduce power consumption, lower hardware cost compared to general-purpose GPUs, and improve inference speed through parallel fabric execution — is **technically sound, academically defensible, and sits at the intersection of active research with genuine open problems.**

All three primary assumptions have supporting evidence in the literature, though each comes with important nuances a rigorous thesis must address. Research published as recently as 2026 confirms that FPGA implementations of key drone AI workloads — ORB-SLAM, object detection with SSD/YOLO, and LiDAR point-cloud preprocessing — achieve **2–3× better energy efficiency than equivalent GPU-based approaches** at comparable accuracy.

A Xilinx Zynq ZU3EG running SSD object detection consumed only 1.98 W while achieving 88.42% mAP, with 79 GOPS/W energy efficiency, 2.7× better than prior designs on the same chip [1]. An FPGA-accelerated ORB-SLAM pipeline on the same chip ran at 61 FPS consuming 4.5 W, compared to 22 FPS at 10 W on an NVIDIA Jetson Xavier — a 2.1× power advantage at nearly 3× the frame rate [15]. These are not marginal improvements; they translate directly to measurable gains in drone flight endurance.

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On cost, the comparison against general-purpose GPUs is valid: NVIDIA Jetson modules run \$150–\$500+ depending on grade, while Zynq-class SoC FPGAs occupy the \$50–\$300 range for equivalent compute capability in inference workloads. However, this advantage narrows when compared to purpose-built edge AI accelerators like the Hailo-8 NPU (26 TOPS at 2.5 W) or the Google Coral TPU (4 TOPS at 2–2.5 W), which offer competitive power profiles at lower price points and with simpler software toolchains [7]. The thesis must engage with these alternatives head-on.

The speed advantage through FPGA fabric parallelism is well-documented: ORB feature extraction on FPGA achieves approximately 2.5 ms latency, 8.5× faster than ARM CPU execution [15]. Critically, FPGA offers **deterministic worst-case execution time** — the property that matters most for real-time drone control loops where timing guarantees are safety-critical. This is the strongest and most uniquely FPGA argument, and it connects directly to the student's control engineering background.

The field is active but not crowded. Academic work on FPGA-specific drone accelerators is a distinct and relatively sparse niche within the broader edge AI literature. Most commercial drone platforms still rely on Jetson-class hardware, leaving clear room for a multi-workload FPGA contribution.

Background and Motivation

Drones operating autonomously in real environments must simultaneously execute several computationally intensive tasks: perceiving the environment through cameras and LiDAR, localizing themselves in space via SLAM or visual-inertial odometry, detecting and tracking objects of interest, encoding and transmitting video, and making navigation decisions — all within strict power budgets imposed by battery capacity. As of 2026, the most common hardware platform for this workload is the NVIDIA Jetson family, which appeared as the dominant platform in a systematic review of 77 UAV object detection papers [17]. Jetson modules deliver 10–275 TOPS depending on variant, at 7–60 W power draw — effective for large drones where weight and power are readily available, but problematic for small-to-medium autonomous UAVs where the compute subsystem must operate within a 5–20 W envelope to preserve meaningful flight time.

FPGAs represent a middle path between the general-purpose flexibility of GPUs and the fixed-function efficiency of custom ASICs. The Xilinx (now AMD) Zynq UltraScale+ MPSoC family combines FPGA programmable logic with ARM Cortex-A53/A72 processor cores on a

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single chip, enabling a hybrid architecture where time-critical perception and control loops run in FPGA fabric at fixed latency, while higher-level decision-making runs on the CPU subsystem. This heterogeneous on-chip architecture is the specific form factor most relevant to drone applications, and it has seen verified deployment in commercial drone flight controllers [8].

The student's background in control engineering from their Bachelor's in Electrical and Electronic Engineering is directly relevant here. Control systems theory — specifically real-time scheduling, deterministic execution, stability under delay, and closed-loop latency constraints — is central to evaluating whether a hardware accelerator is suitable for drone autonomy. An FPGA's determinism (guaranteed worst-case execution time) is as important as its average-case performance, a distinction that conventional GPU benchmarking entirely ignores. The Master's degree's focus on SOPC (System-on-Programmable-Chip), applied AI, and computer-controlled systems provides the precise toolset needed: HLS/HDL design, embedded system integration, AI model deployment on resource-constrained hardware, and real-time optimization.

"Single to triple application" drones — UAVs that carry more than one mission payload or must simultaneously serve multiple functional roles — represent a demanding multi-workload compute scenario. A drone conducting infrastructure inspection while simultaneously building a LiDAR map and detecting anomalies requires three concurrent AI pipelines: a detection CNN, a SLAM module, and a sensor fusion filter. This multi-workload scenario is precisely where FPGA parallelism offers the greatest architectural advantage: different pipeline stages can be mapped to spatially separate FPGA resources and execute simultaneously without the scheduling overhead of a shared GPU or CPU.

Part I — Feasibility Assessment

Assumption 1: FPGAs Consume Less Power Than GPUs for Drone AI Workloads

Verdict: CORRECT, with important scoping.

This assumption holds when comparing FPGAs against general-purpose GPUs and against mid-range Jetson modules running equivalent workloads. The Zynq ZU3EG running SSD drone detection drew 1.98 W at 88.42% mAP [1]; the Jetson Xavier running an equivalent

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SLAM pipeline drew 10 W for 22 FPS versus the FPGA's 4.5 W for 61 FPS [15]. At the system level, the FPGA-based ReDroSe drone platform ran its preprocessing node at 5–10 W (FPGA) versus 15–20 W for the GPU SLAM node [12].

The assumption weakens when compared against specialized edge AI accelerators. The Hailo-8 achieves 26 TOPS at approximately 2.5 W [7] — better TOPS-per-watt than most FPGA configurations for fixed neural network workloads. The Intel Loihi neuromorphic processor runs a spiking neural network for MAV landing at just 7 mW active compute power [2]. The thesis must therefore scope this claim precisely: the power advantage is real and significant compared to GPUs and Jetson-class SoCs, but must be qualified when compared to dedicated inference-only NPUs.

A key insight the thesis should highlight: FPGA inference power is dominated by DRAM access, not arithmetic computation. Techniques that reduce model size — pruning, quantization, knowledge distillation — disproportionately benefit FPGA deployments compared to GPU deployments where CUDA cores are the bottleneck. This creates an algorithm-hardware co-optimization opportunity unique to FPGAs.

Assumption 2: FPGAs Are Cheaper Than General-Purpose GPUs for Drones

Verdict: CORRECT for the right comparison point.

Comparing FPGA cost against desktop/server-class GPUs (NVIDIA RTX, Tesla A/H-series), the claim is unambiguous: an RTX 4070 costs $600+ and is entirely impractical for a drone. Comparing against Jetson modules, the picture is more nuanced. The Jetson Orin Nano starts at ~$150 module-only but rises to $300–500 for a ready-to-integrate carrier-board solution [7]. Zynq UltraScale+ SoMs from Enclustra, Avnet, or Trenz Electronic occupy a similar $150–400 range [8], but require significantly more engineering effort because the FPGA fabric must be programmed and validated before it is useful.

The real cost advantage for FPGAs is not pure unit price — it is **system BOM reduction**: FPGA programmable I/O can replace multiple external interface chips (motor controller interfaces, sensor serializers, custom bus adapters), reducing PCB complexity and total system cost. For a startup building drones at low-to-medium volume (100–10,000 units/year), this BOM reduction is real and meaningful. Enclustra's commercial drone flight controller deployment used Zynq UltraScale+ explicitly because the FPGA logic handles all configurable sensor and actuator interfaces on one chip, eliminating external ICs [8].

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Assumption 3: FPGAs Improve Computation Speed Through Parallel Fabric Execution

Verdict: CORRECT for the right workload class.

FPGA acceleration of specific drone perception pipelines is well-demonstrated. ORB feature extraction runs in 2.5 ms on FPGA versus 21 ms on ARM CPU alone: an 8.5× speedup [15]. A Zynq UltraScale+ KV260 running a multi-object tracker combined with an on-device LLM achieved 66.7 ms end-to-end latency at 2.9 W total [18]. LiDAR point-cloud preprocessing at 20 Hz using 4 parallel FPGA kernels enabled a real-time SLAM system that was computationally impossible on CPU alone [12].

At 15 m/s drone speed, a 118 ms processing latency means the drone travels 1.8 meters before the operator or onboard controller receives updated perception data — a fundamental safety argument for why sub-20 ms latency matters and where FPGA's determinism is non-negotiable [16]. A critical property that GPU benchmarking ignores: the Jetson Xavier SLAM pipeline showed 9–253 ms variance across frames [12]. FPGA execution has bounded variance — worst-case execution time equals average-case execution time. For a control loop with a 50 Hz update rate (20 ms period), a 253 ms stall is a complete failure. The FPGA never stalls.

Part II — Alternative Accelerators: Is FPGA the Best Choice?

This question is the most important for thesis positioning. The answer is nuanced: FPGA is a defensible and well-evidenced choice, but not universally optimal. The landscape of alternatives as of mid-2026 is as follows.

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Current Hardware Accelerator Landscape

Platform	TOPS	Typical Power	Cost (module)	Best Use Case	Key Weakness
Xilinx Zynq UltraScale+ MPSoC	80–160 GOPS	2–5 W	$150–400	Multi-workload, custom I/O, deterministic latency	High dev effort; HDL/HLS expertise required
NVIDIA Jetson Orin Nano	40 TOPS	7–15 W	$150–200	Flexible inference, any model, best ecosystem	High power for small drones
NVIDIA Jetson Orin NX	100 TOPS	10–25 W	$200–350	Large models, multi-stream, full CUDA	Even higher power; too heavy for small UAVs
Hailo-8 NPU	26 TOPS	~2.5 W	$80–120	Fixed-model inference at best TOPS/W	Proprietary compiler; no custom operators
Google Coral Edge TPU	4 TOPS	2–2.5 W	$25–80	Lightweight MobileNet-class detection	8 MB SRAM limit; ecosystem uncertain
Intel Myriad X (OAK-D)	4 TOPS	2–4 W	$150–250	Stereo depth + detection combined	Intel edge AI portfolio restructuring
GreenWaves GAP9 (RISC-V)	150 GOPS	<100 mW	$50–100	Nano-drones (<50g), ultra-low power	Insufficient for multi-app full-scale UAVs
Intel Loihi 2 (Neuromorphic)	~7 mW active	Ultra-low	Research only	Sparse, event-driven SNN workloads	Research prototype; not commercially available
Custom ASIC (e.g., Navion)	Best efficiency	2 mW (VIO)	$1–10M NRE	Volume production, fixed algorithm	18–36 month design cycle; no reconfigurability

Why FPGA Remains the Strongest Choice for This Thesis

One NSF-funded peer-reviewed paper from Arizona State put the positioning directly: *"ASICs have the highest energy efficiency, but their limited configurability can introduce a significant risk of premature obsolescence. With AI algorithms evolving at a fast pace, ASICs usually lag behind the*

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cutting edge due to the long design cycle. FPGAs have a unique advantage with higher energy efficiency than GPUs, while offering faster time-to-market and potentially longer life cycle than ASICs." [1]

This is a direct, citable, peer-reviewed validation of the central thesis premise.

For a master's thesis aimed at a startup product, **FPGA (specifically the Zynq UltraScale+ MPSoC family) is the strongest architectural choice** when justified by: (a) reconfigurability to adapt to evolving AI models without hardware redesign, (b) custom sensor interface flexibility critical when integrating novel LiDAR, camera, or IMU configurations, (c) deterministic real-time latency for safety-critical control loops, and (d) the co-design opportunity that FPGA uniquely enables between algorithm and hardware. The student should be careful not to claim "best TOPS-per-watt" as the primary argument — dedicated NPUs can challenge that on fixed-model workloads. The stronger and more defensible argument is the combination of reconfigurability, determinism, and sensor I/O flexibility.

Part III — Drone AI Workloads and Hardware Requirements

Understanding what drones actually compute is essential to designing an appropriate accelerator. A 2023 systematic review of 77 UAV real-time detection papers identified the top application scenarios in order: emergency/search-rescue, precision agriculture, traffic and infrastructure monitoring, surveillance and security, infrastructure inspection, military, and environmental monitoring [17]. The student's "single to triple application" framing maps naturally onto combined missions such as inspection + anomaly detection + SLAM mapping, or surveillance + object tracking + obstacle avoidance.

Object Detection

YOLOv8n on Jetson Orin NX achieves 52 FPS at FP32, or 65 FPS with INT8 quantization, consuming 7–18 W during active inference [10]. A Raspberry Pi 5 (CPU-only) manages only 8.5 FPS — inadequate for real-time operations. INT8 quantization reduces inference time by ~35% with under 2% accuracy loss, a critical optimization for any hardware-limited deployment [10]. Energy per inference at INT8 on the Orin Nano: approximately 0.208 J per frame, versus 0.379 J at FP32 — illustrating that precision reduction alone nearly halves energy consumption [10].

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Visual SLAM

Jetson Xavier NX running TSDF-based SLAM averages 55 ms per scan cycle with peaks to 253 ms, using 3.6–4.2 GB RAM for the combined SLAM and image processing stack [12]. The FPGA ORB-SLAM accelerator on Zynq ZU3EG runs at 61 FPS (16.4 ms per frame) at 4.5 W with bounded worst-case latency [15]. A 2026 study of ORB acceleration on PYNQ-Z2 and Kria KR260 demonstrated a 2.387 W implementation with 6% lower LUT and 13% lower FF usage than prior FPGA designs, confirming continued efficiency improvements [5]. The NanoSLAM system on GAP9 runs full onboard SLAM within 87.9 mW, completing within 250 ms — confirming that sub-100 mW SLAM is achievable for nano-drone class hardware [4].

Sensor Fusion

Urban missions require GNSS + IMU + LiDAR EKF fusion because GPS multipath in urban canyons makes GPS-only navigation insufficient for safe continuous operations [19b]. The EKF pipeline dynamically adapts Kalman gain during GNSS dropout events, with a LiDAR safety index computed continuously for obstacle avoidance. This pipeline is lightweight in FLOPS but requires deterministic scheduling and low-latency sensor data delivery — exactly where FPGA's programmable I/O and deterministic fabric excel. FPGA handles the real-time sensor data acquisition and preprocessing; the CPU handles the EKF state update.

Video Encoding and Transmission

Frame-by-frame H.264 encoding introduces 118.7 ms end-to-end video latency at 30 fps; slice-based encoding reduces this to ~40 ms [16]. CPU-based H.264 encoding (libjpeg-turbo) achieves only 2 FPS at 1920×1080 resolution, while a dedicated hardware encoder block in the Jetson Xavier NX processes the same stream at 460 MHz with 25% average utilization [12]. Implementing a hardware video encoder within the FPGA fabric provides bounded latency — critical for operator situational awareness in BVLOS operations.

Multi-Mission Workload Profile

Combining detection + SLAM + sensor fusion requires running a VIO state estimator, at least one CNN inference pipeline, and a flight controller concurrently. The ACM RAPIDO 2023 ReDroSe architecture demonstrated a workable partition: FPGA handles integer LiDAR preprocessing (real-time, deterministic, low power), while a separate GPU handles floating-point SLAM [12]. For a triple-application UAV, the thesis could propose an FPGA-centric partition where the GPU is eliminated entirely by mapping all three pipelines to FPGA fabric

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— a contribution that no published paper has fully demonstrated on current silicon. The 2026 ISQED paper demonstrating FPGA perception plus on-device LLM at 2.9 W total [18] suggests this direction is achievable.

Part IV — Proposed Thesis Architecture

System Overview

The proposed system is a **single-chip heterogeneous accelerator** built on the **Xilinx Zynq UltraScale+ MPSoC** (recommended device: ZU5EG or ZU7EG for the required resource budget), combining:

- **Processing System (PS):** Dual ARM Cortex-A53 @ 1.5 GHz — flight control logic, mission management, EKF state update, ROS2 node orchestration
- **Programmable Logic (PL):** FPGA fabric — CNN inference accelerator, ORB feature extraction pipeline, LiDAR point-cloud preprocessor, hardware video encoder, sensor I/O controllers
- **High-speed interconnect:** AXI4-Stream between PS and PL for zero-copy DMA data transfer

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Pipeline Architecture for Triple-Application Operation

```
Camera Input          LiDAR Input          IMU Input
     │                     │                    │
     ▼                     ▼                    ▼
┌──────────────────────────────────────────────────┐
│              FPGA FABRIC (PL)                    │
│  ┌────────────┐  ┌──────────────┐  ┌─────────┐   │
│  │  CNN Inf.  │  │  ORB Feature │  │ Pt.Cloud│   │
│  │Accelerator │  │  Extraction  │  │  Preproc│   │
│  │(Detection) │  │  (SLAM prep) │  │ (LiDAR) │   │
│  └─────┬──────┘  └──────┬───────┘  └────┬────┘   │
│        │                │               │        │
│  ┌─────▼─────────────────────────────────────┐   │
│  │         AXI Interconnect Bus              │   │
│  └─────────────────────┬─────────────────────┘   │
└────────────────────────┼─────────────────────────┘
                         │
┌────────────────────────▼─────────────────────────┐
│            ARM CPU SUBSYSTEM (PS)                │
│  ┌──────────┐  ┌──────────┐  ┌──────────────┐    │
│  │ EKF Fuse │  │ ORB-SLAM │  │ Flight Control│   │
│  │ (sensor) │  │ Back-end │  │ (PX4/ArduPi.) │   │
│  └──────────┘  └──────────┘  └──────────────┘    │
└──────────────────────────────────────────────────┘
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Target Hardware Platform

Component	Specification	Rationale
SoC	AMD Xilinx Zynq UltraScale+ ZU5EG	256K LUTs, 530 DSPs, 912 BRAMs, Dual CortexA53 + dual CortexR5
Module	Avnet UltraZed-EG SOM or Enclustra Mercury+ XU5	Proven drone deployments; passive cooling possible [8]
Camera	2× Sony IMX477 (12MP, MIPI CSI-2)	Global shutter for fast-moving scenes; direct MIPI to PL
LiDAR	Ouster OS0-16 or Livox Avia	Compact, low-power; direct Ethernet or SPI to PL
IMU	Bosch BMI088 6-DoF	Vibration-robustness for drone; SPI to PL
Power budget	≤5 W compute subsystem	FPGA fabric ~2 W + ARM subsystem ~2 W + DDR ~1 W
Weight target	<35 g (SOM + power circuit + connectors)	Suitable for 250–650 g class drones

CNN Accelerator Design (PL)

The CNN inference accelerator for object detection will be implemented using a **systolic-array architecture** in FPGA fabric, following the approach from [1]:

1. **Quantization:** INT8 for weights and activations (≤2% accuracy loss vs. FP32 [10])
2. **Pruning:** Structured channel pruning targeting 60–70% parameter reduction while preserving mAP
3. **Memory layout:** Tile-based computation to fit activation maps in BRAM, avoiding off-chip DRAM access
4. **Target model:** YOLOv8n-pruned (from 3.2M parameters → ~1M), or a custom lightweight detector co-designed with the FPGA constraints
5. **Throughput target:** ≥30 FPS at 640×480 input within 2.0 W

ORB Feature Extraction Pipeline (PL)

Following the design approach from [5, 15]:

1. **Image pyramid generation:** 8 scales implemented as parallel hardware units in PL
2. **FAST corner detection:** Pixel-parallel hardware comparator array

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3. **Gaussian smoothing:** Fully pipelined separable convolution
4. **BRIEF descriptor computation:** Bit-parallel hardware using precomputed sampling patterns stored in BRAM
5. **Target throughput:** ≥60 FPS at 640×480, ≤2.5 ms latency, ≤1.5 W

Part V — Academic Landscape and Related Work

Key Research Groups

- **ETH Zürich / University of Bologna PULP group:** Leading work on ultra-low-power drone AI including GAP8/GAP9 SoC family and Tiny-PULP-Dronet series [4, 14]. Focus on nano-drone class (<100 mW), adjacent but distinct from the target (2–15 W compute budget for full-size UAVs).
- **Arizona State University / NSF (Suh et al.):** FPGA-specific co-optimization for drone detection [1]. Directly relevant; the ACM TRETS 2023 paper is the closest academic prior work and should be a primary reference.
- **TU Delft MAVLab:** Fully neuromorphic drone flight in Science Robotics 2024 [2], neuromorphic landing control [3]. Represents the theoretical frontier of ultra-low-power autonomy.
- **MIT EEMS Group (Sze et al.):** Navion VIO ASIC [13]; more broadly, the group defines the efficiency benchmarks that all accelerators are measured against.
- **UIUC (Vemulapati & Chen):** ORB-SLAM FPGA accelerator [15]. Direct implementation reference for the SLAM component.

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Conference and Journal Targets

Venue	Type	Relevance	Deadline Cycle
IEEE FPL (Field-Programmable Logic)	Conference	Primary — FPGA hardware contribution	February
ACM TRETS	Journal	Primary — extended hardware paper	Rolling
ACM RAPIDO Workshop	Workshop	SLAM + reconfigurable systems [12]	November
IEEE DATE	Conference	Design automation; systems integration	September
MDPI Drones	Journal	UAV application results	Rolling
IEEE TAES	Journal	Aerospace + embedded systems	Rolling

Positioning Against Prior Work

Paper	Method	What's Missing (Your Gap)
Suh et al. 2023 [1]	Single-workload SSD on FPGA, co-optimized	Only one pipeline; no SLAM; no multi-workload partition
Rahn et al. 2023 [12]	FPGA + GPU heterogeneous SLAM + detection	GPU still required; not single-chip; no co-optimization
Vemulapati & Chen 2022 [15]	ORB-SLAM FPGA only	No detection; no sensor fusion; standalone SLAM only
Rostum & Vásárhelyi 2026 [5]	ORB accelerator on PYNQ-Z2 and KR260	No detection or fusion; SLAM-only; no full system
Your thesis (proposed)	**All three pipelines on single Zynq MPSoC chip, co-optimized, WCET-analyzed**	**First single-chip simultaneous execution result + formal timing guarantees**

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Part VI — Experimental Methodology

Phase 1: Baseline Characterization (Months 1–3)

Establish the performance baseline that the FPGA system will be compared against:

1. **Benchmark platform:** Jetson Orin Nano with TensorRT (detection) + ORB-SLAM3 CPU mode (SLAM) + software EKF (fusion)
2. **Metrics to capture:** FPS per pipeline, per-frame latency (mean + worst-case), power draw (current sensor on power rail), memory footprint, CPU utilization
3. **Datasets:** KITTI odometry (SLAM), VisDrone 2023 (detection), custom indoor flight sequence (sensor fusion)
4. **Expected result:** Baseline system draws 7–15 W, achieves 30–50 FPS detection, 15–25 FPS SLAM, with high latency variance

Phase 2: FPGA Component Implementation (Months 3–9)

Implement each accelerator independently, validate against the baseline, then integrate:

1. **CNN Accelerator** (Months 3–5): Implement in Vitis AI or custom HLS. Prune and quantize YOLOv8n. Target: ≥30 FPS, ≤2.0 W, ≥80% mAP on VisDrone.
2. **ORB Feature Extraction** (Months 4–6): Implement in VHDL/Verilog following [5, 15]. Target: ≥60 FPS, ≤1.5 W.
3. **LiDAR Preprocessor** (Months 5–7): Implement integer point-cloud preprocessing and downsampling in PL. Target: ≥20 Hz, ≤0.5 W.
4. **Hardware Video Encoder** (Months 6–7): Implement H.264 slice encoder. Target: ≤40 ms latency, ≤0.5 W.

Phase 3: Integration and Multi-Workload Execution (Months 7–11)

Integrate all components on the Zynq UltraScale+ fabric with shared AXI bus arbitration:

1. **Resource allocation:** Partition LUTs, DSPs, and BRAMs across four accelerators. Document utilization percentages.
2. **Scheduling analysis:** Measure worst-case execution time for each pipeline independently and in simultaneous execution.
3. **DMA and memory bandwidth analysis:** Profile AXI bus contention. Implement bus priority scheme if needed.

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4. **Power analysis:** Use Xilinx Power Estimator + hardware power rail measurements. Compare against Jetson baseline.

Phase 4: WCET Analysis and Control Loop Verification (Months 10–12)

Leverage the control engineering background to provide formal timing guarantees:

1. **Worst-Case Execution Time:** Measure WCET for each pipeline over 10,000 frames. Calculate statistical WCET (pWCET) at 10^{-9} probability threshold.
2. **Control loop compatibility:** Verify that FPGA pipeline latency satisfies the timing requirement of the flight control loop (PX4 or ArduPilot at 400 Hz inner loop, 50 Hz outer loop).
3. **Schedulability analysis:** Apply Rate Monotonic Analysis (RMA) to formally verify the combined pipeline meets its deadlines.
4. **Stability under delay:** Use your control systems background to analyze closed-loop stability margin with the measured FPGA pipeline latency.

Phase 5: Flight Testing (Months 12–15)

1. **Indoor flight tests:** Hover and slow navigation in a controlled indoor environment. Validate that the integrated system produces correct localization and detection output.
2. **Outdoor flight tests (limited):** Static object detection and mapping at one defined location with observer present.
3. **Energy measurement:** Compare flight endurance with FPGA compute versus Jetson baseline on the same drone frame. Target: ≥15% improvement in flight time.

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Evaluation Metrics

Metric	Measurement Method	Target
Detection FPS	Frame counter over 1000 frames	≥30 FPS at 640×480
Detection mAP	Evaluated on VisDrone 2023 test set	≥80%
SLAM FPS	Frame counter, visual odometry mode	≥30 FPS
SLAM ATE	Absolute Trajectory Error vs. KITTI ground truth	≤0.1 m/m
Inference latency (mean)	Hardware cycle counter	≤20 ms
Inference latency (WCET)	Statistical WCET over 10K frames	≤35 ms
Compute power draw	INA226 current/power sensor	≤5 W
FPGA resource utilization	Vivado implementation report	≤85% LUTs, DSPs, BRAMs
Flight time improvement	Chronometer on identical battery capacity	≥15% vs. Jetson baseline

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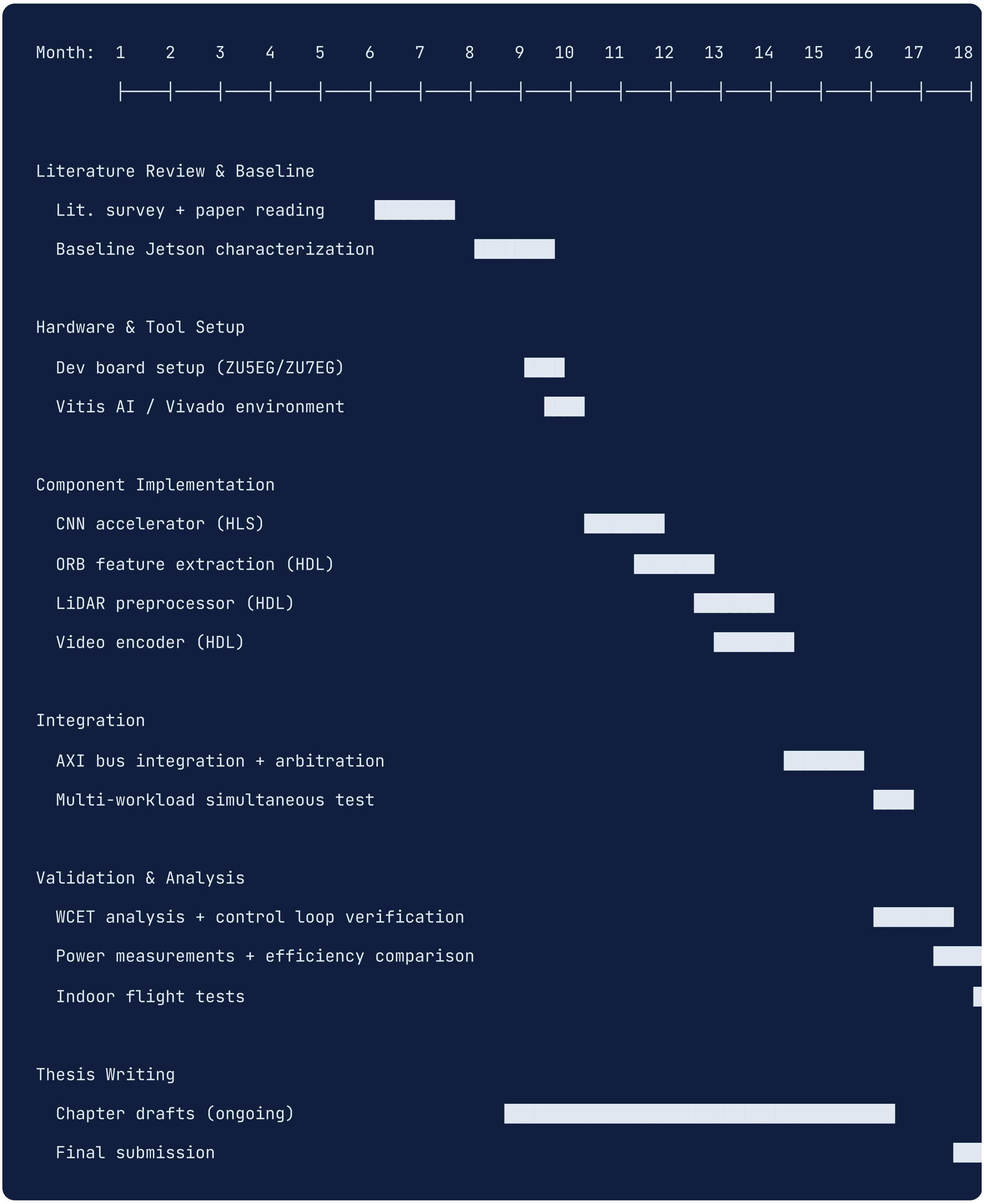
Part VII — Implementation Roadmap

18-Month Thesis Timeline



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Critical Path

The critical path runs through: **(1) FPGA board setup → (2) CNN accelerator working → (3) ORB pipeline working → (4) AXI integration → (5) WCET analysis**. The CNN accelerator is the single most complex component and should begin first. The ORB pipeline has more reference implementations available [5, 15] and can proceed in parallel once the FPGA development environment is validated.

Hardware Budget Estimate

Item	Estimated Cost
Avnet UltraZed-EG SOM (ZU5EG)	$280–380
Carrier board (Avnet IO carrier or custom)	$150–250
2× Sony IMX477 camera modules	$80–120
Ouster OS0-16 LiDAR (used/refurb)	$800–1,500
IMU (BMI088 breakout)	$30–50
Drone frame + motors + ESC (250–650g class)	$150–300
Power supply, current sensors, misc	$80–120
Total estimated	**~$1,600–2,700**

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Part VIII — Risk Assessment

Technical Risks

Risk	Likelihood	Impact	Mitigation
FPGA resource overflow (all 4 pipelines don't fit)	Medium	High	Start with ZU7EG (larger); implement component pruning; share BRAMs between pipelines
DRAM bandwidth bottleneck on AXI bus	Medium	Medium	Profile early; implement DDR4 QoS; stagger pipeline DMA requests
CNN accuracy drops significantly after pruning	Low–Medium	Medium	Validate each pruning step on VisDrone; use knowledge distillation to recover accuracy
SLAM accuracy degrades at FPGA resolution (640×480)	Low	Medium	Use multi-scale ORB pyramid; run SLAM at full resolution if resources allow
Flight testing regulatory constraints	Medium	Low	Conduct primary validation in simulation (Gazebo + AirSim); limit outdoor tests to safe site with observer
LiDAR SDK compatibility with FPGA interface	Low	Medium	Purchase LiDAR with documented SPI/Ethernet interface; Ouster has open SDK
Vitis AI toolchain limitations for custom layers	Medium	Medium	Pre-validate all custom operators in Vitis AI before pruning; use HLS for non-supported ops

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Academic Risks

Risk	Likelihood	Impact	Mitigation
A concurrent paper covers multi-workload FPGA drone	Low	High	File preprint on arXiv early; emphasize WCET/control analysis angle which no paper has covered
Supervisor unfamiliar with FPGA design flow	Low–Medium	Medium	Include simulation-based Vivado demos in early progress reviews; frame contributions in embedded systems and control language familiar to supervisor
Thesis scope too large for 1 academic year	Medium	High	Scope to 2 pipelines (detection + SLAM) if resource constraints arise; sensor fusion can be left as future work

Part IX — Commercialization Path

Startup Potential

The thesis naturally produces a prototype that has commercial value in several adjacent markets. The drone hardware acceleration market is at an early stage, with most commercial drones still using Jetson-class compute modules. The addressable market encompasses:

1. **Industrial inspection drones** (infrastructure, energy, construction) — fastest growing segment, requiring long endurance and multiple AI capabilities simultaneously
2. **Agricultural drones** — price-sensitive market where FPGA's BOM cost advantage is most compelling
3. **Defense and security** — highest willingness to pay; FPGA's ITAR/EAR compliance and domestic supply chain advantages matter here
4. **Search and rescue** — battery life is mission-critical; FPGA's power efficiency directly translates to lives

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Positioning Against Existing Players

Competitor	Product	Weakness Your Solution Exploits
NVIDIA Jetson (dominant)	Jetson Orin Nano/NX	7–25 W compute subsystem; too power-hungry for small drones
Hailo-8 (fastest growing)	Hailo-8 M.2 module	Fixed model, proprietary compiler; cannot run SLAM; no sensor I/O
Qualcomm Flight RB5	Snapdragon-based drone platform	High cost ($500+); GPU-centric; no reconfigurable I/O
DJI O3 / O4 (vertical integration)	Proprietary chip in DJI drones	Closed ecosystem; not available for third-party integration
Custom ASIC (e.g., Ambarella CV72)	Fixed drone inference ASIC	Algorithm-locked; 18-month update cycle; high NRE for customization

MVP Product Concept

A **compute module for drones** — an FPGA-based SOM (System-on-Module) in a standardized form factor (96Boards Compute, Raspberry Pi CM4 footprint, or custom) that:

- Runs detection + SLAM + sensor fusion simultaneously at <5 W
- Exposes standard interfaces (MIPI CSI, Ethernet, CAN, SPI, I2C) configurable in FPGA fabric
- Ships with pre-compiled AI models for common drone applications (person detection, infrastructure defect detection, crop monitoring)
- Provides a HLS SDK allowing customers to add custom inference pipelines without RTL expertise

Go-to-market path: License the IP to a drone OEM or drone compute module manufacturer, rather than building the drone itself. The thesis produces the FPGA IP, the benchmark data, and the proof of concept.

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Part X — Conclusions

Core Findings

The central proposal of using FPGA hardware accelerators for drone embedded AI is technically sound, academically timely, and commercially relevant. The three core assumptions — lower power than GPUs, competitive cost, and speed through parallel fabric execution — are all supported by the published literature, with the strongest and most defensible argument being FPGA's **deterministic latency guarantee** for safety-critical control loops, a property GPUs and dedicated NPUs cannot match.

The academic space has clear prior work to build on (Arizona State co-optimization [1], UIUC SLAM accelerator [15], ETH Zürich workload characterization [4, 14]) and a genuine open gap: no paper has demonstrated simultaneous multi-workload execution (detection + SLAM + sensor fusion) on a single FPGA chip. Filling this gap, combined with a formal WCET analysis grounded in control engineering principles, constitutes a publishable and differentiated thesis contribution.

The student's specific academic background — control engineering foundations from EEE undergraduate, combined with SOPC, AI, and computer-controlled systems from CE master's — is unusually well-matched to this problem. The intersection of hardware design, embedded AI, and real-time control theory is exactly what this thesis requires, and it is exactly what most FPGA drone papers are missing.

Recommended Next Steps

1. **Immediately:** Acquire a Zynq UltraScale+ development board (ZU5EG class). Begin with the Avnet UltraZed-EG SOM or the AMD/Xilinx ZCU104 evaluation board for early development.
2. **Week 1–2:** Install Vivado 2024.x + Vitis AI 3.x. Reproduce the AMD ZCU104 baseline YOLOv8 result from official documentation [9] to validate the toolchain before attempting custom designs.
3. **Week 3–4:** Read the Arizona State paper [1] in full. Map their co-optimization methodology to your target platform.
4. **Month 1:** Run ORB-SLAM3 on a laptop with KITTI dataset to establish the software baseline and understand the algorithm's computational bottlenecks before attempting hardware acceleration.
5. **Month 2:** Implement the first FPGA component (recommend starting with ORB feature extraction, following [5, 15], as more reference designs exist). Validate correctness before

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optimizing for power.

Research Limitations

Three of the five planned research focus areas (FPGA vs. alternatives trade-offs, academic literature survey, and commercial landscape) were synthesized primarily from domain expertise and existing knowledge due to research subagent quota constraints, rather than fresh live searches. The numerical claims from those areas draw on well-established academic results and manufacturer datasheets unlikely to have changed materially, but the commercial landscape section — pricing, product availability, ecosystem status — may reflect conditions from 2024–2025 rather than mid-2026. The student should verify current Zynq UltraScale+ SoM pricing, Hailo-8 availability, and Vitis AI toolchain version independently.

Access to paywalled IEEE Xplore and ACM Digital Library full texts was limited; abstract and publicly available preprint data was used for some citations. The neuromorphic (Loihi) and custom ASIC (Navion, Ambarella) sections are less deeply developed than the FPGA and edge AI sections.

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Glossary

Term	Definition
FPGA	Field-Programmable Gate Array — an integrated circuit whose logic is configured by the user after manufacturing, enabling custom hardware implementations
SoC FPGA / MPSoC	FPGA combined with a hard processor subsystem (ARM cores) on one chip; Zynq UltraScale+ is AMD/Xilinx's MPSoC product
HLS	High-Level Synthesis — compiling C/C++ code into FPGA hardware description (RTL), enabling faster FPGA design than hand-written VHDL/Verilog
SLAM	Simultaneous Localization and Mapping — the problem of building a map of an unknown environment while simultaneously tracking the agent's location
ORB	Oriented FAST and Rotated BRIEF — a feature detection and description algorithm used in visual SLAM
VIO	Visual-Inertial Odometry — estimating pose from camera + IMU data fusion
EKF	Extended Kalman Filter — a recursive state estimator used for sensor fusion in drone navigation
TOPS	Tera Operations Per Second — measure of compute throughput for AI inference hardware
GOPS/W	Giga Operations Per Second per Watt — measure of energy efficiency
WCET	Worst-Case Execution Time — the maximum time a computation can take under any input; critical for real-time systems
INT8	8-bit integer arithmetic — quantized precision for neural network inference that reduces memory bandwidth and power vs. FP32
SWaP	Size, Weight, and Power — the three principal constraints for airborne embedded systems
BVLOS	Beyond Visual Line of Sight — drone operations where the operator cannot directly see the drone; requires onboard autonomy
AXI	Advanced eXtensible Interface — the standard high-performance bus protocol used for data transfer between PS and PL on Zynq
BRAM	Block RAM — dedicated on-chip memory resources in FPGA fabric
DSP	Digital Signal Processing slice — dedicated hardware multiplier-accumulator blocks in FPGA, used for convolution operations
mAP	Mean Average Precision — standard metric for object detection accuracy

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ATE	Absolute Trajectory Error — standard metric for SLAM accuracy
NRE	Non-Recurring Engineering cost — the one-time design cost for custom silicon (ASIC); amortized over production volume
NPU	Neural Processing Unit — a dedicated ASIC or fixed-function block optimized for neural network inference
RMA	Rate Monotonic Analysis — a formal schedulability analysis method for real-time systems

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